
Nikolaos Bakirtzis

Software Developer

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SUMMARY

Portfolio Website: <https://nnbakirtzis.github.io/Portfolio-Website/>

Software Engineer with 2 years of full-time experience building production applications, automation systems, data pipelines, and operational infrastructure. Experienced with Python, TypeScript/JavaScript, SQL, Linux, APIs, and database-backed systems, with hands-on experience building LLM and agentic AI applications. Strong track record of solving ambiguous technical problems and owning systems from architecture through production deployment.

EDUCATION

Ohio Wesleyan University, Delaware, OH - *BSc in Computer Science, Minor in Business*

August 2020 - May 2024

Skills: C++, Python, HTML, CSS, Assembly, Operating Systems

SKILLS

- **Technical Skills:** Python, Linux, Information Security, TypeScript/JavaScript, SQL,
- **AI:** LLM APIs, AI Agents, Agentic Software Development, RAG Applications, Prompt Engineering
- **Frameworks:** FastAPI, React, Next.js, Angular, React Native
- **Data:** PostgreSQL, MySQL, SQLite, InfluxDB, Supabase
- **Infrastructure:** Linux, Windows Server, Docker, Git, Grafana, REST APIs

CERTIFICATIONS

- Post Graduate Program in Generative AI for Business Applications (Texas McCombs School of Business)
- Palo Alto Networks Cybersecurity Specialization (Palo Alto Networks)
- Google Cybersecurity Specialization (Google)

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- Ethical Hacking from Scratch (zSecurity)
 - HackTheBox (HTB) - Practical experience in penetration testing, Kali Linux toolkit, and vulnerability exploitation by solving security challenges on virtual machines.

WORK EXPERIENCE

University of Texas at Austin, US - *Software Developer*

August 2024 - Present (Full Time)

- In charge of developing and maintaining production scientific applications and data systems supporting TexNet's statewide seismic monitoring infrastructure. Open-source scientific applications include:
 - https://github.com/ut-beg-texnet/fault_slip_potential
 - <https://github.com/ut-beg-texnet/GIST/tree/main>
- Built Python automation and data-processing pipelines for backfilling datasets, health monitoring, log analysis, validation, and operational data collection, reducing repetitive manual engineering work for over 100+ seismic stations.
- Designed and implemented real-time observability and station health systems using Python, InfluxDB, SQL databases, Grafana, Linux services, and custom data-ingestion pipelines.
- Modernized and optimized scientific web applications, improving web-based workflows, performance by roughly 5x, visualization, and usability.
- Build and maintain dashboards, scheduled services, and internal engineering tooling across Linux and Windows production environments.
- Diagnose production issues across applications, databases, networks, services, and data pipelines, collaborating directly with scientists and technical stakeholders to translate domain requirements into software solutions.

Ohio Wesleyan University, US - *Cybersecurity Analyst*

September 2023 - May 2024 (Part Time)

- Led the implementation of the NIST 800-171 framework, directly enhancing threat detection and incident response capabilities.
- Orchestrated comprehensive internal and external penetration tests to identify and mitigate vulnerabilities, leading to the development and implementation of robust security measures that reduced potential attack vectors.

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- Managed weekly meeting agendas and coordinated with the IT team on security control implementations, showcasing project management and communication skills.

Ohio Wesleyan University, US - *Computer Science Lab Assistant*

January 2023 - May 2023 (Part Time)

- Supported 20+ students in understanding complex programming concepts, highlighting an ability to communicate technical information effectively.

Vidavo, Thessaloniki, Greece - *Front-End Web Developer*

June 2022 - August 2022 (Internship)

- Developed and enhanced responsive web designs, contributing to a broader understanding of IT architecture using HTML5, Bootstrap, and CSS.

Projects

AI Legal Research Assistant

Tech Stack: Python, FastAPI, LLM APIs, RAG, Web Search, React/Next.js

- Built an AI assistant for Greek legal professionals combining LLM reasoning (Gemini/Deepseek) with authoritative legal-source retrieval and citation-backed answers.
- Implemented research workflows, citation verification, retrieval evaluation, model fallback, and structured responses.
- Designed the system to reduce hallucinations by grounding responses in trusted legal sources rather than relying solely on model knowledge.

Multi-Agent Career Intelligence Platform

Tech Stack: React Native, Supabase, LLM APIs, Agentic Workflows

- Designed a multi-agent AI system coordinating specialized agents for job matching, labor-market analysis, compensation forecasting, and career planning.
- Built structured agent handoffs, shared context, user-state persistence, and model-driven recommendations.